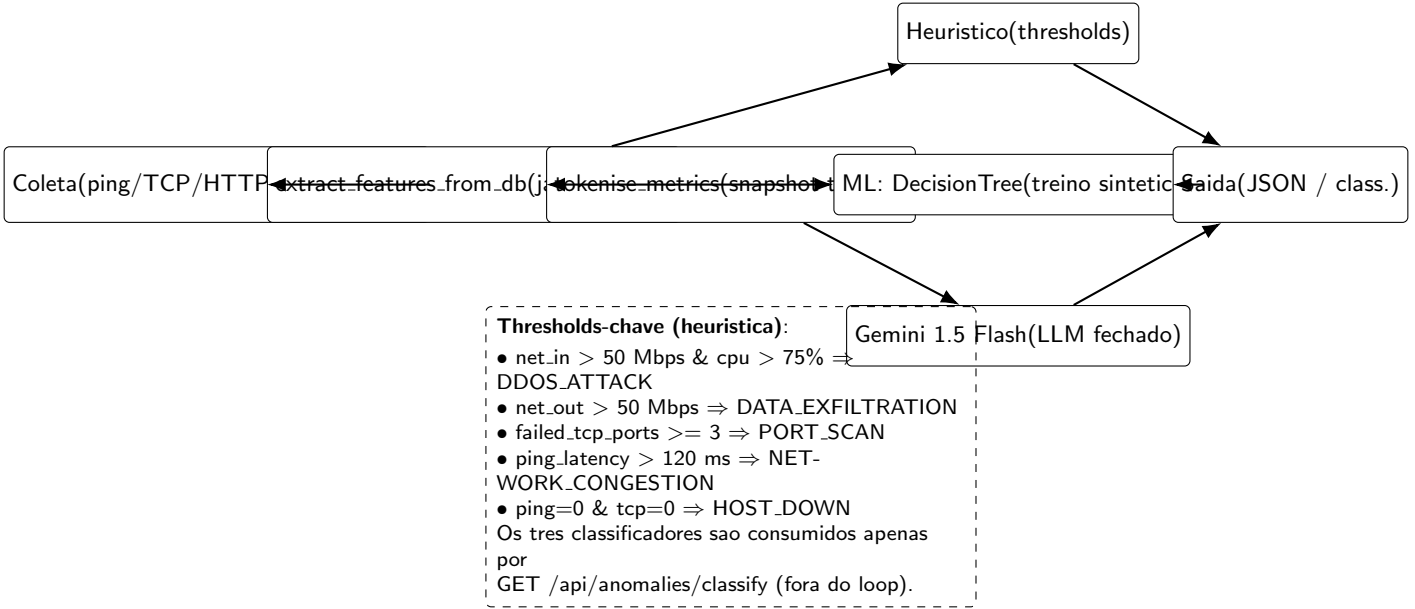

Heuristico(thresholds)
Coleta(ping/TCP/HTTP
extract_features_from_db(j
tokenise_metrics(snapshot
ML: DecisionTree(treino sintetic
Saida(JSON / class.)
Gemini 1.5 Flash(LLM fechado)
Thresholds-chave (heuristica):
• net_in > 50 Mbps & cpu > 75% ⇒ DDOS_ATTACK
• net_out > 50 Mbps ⇒ DATA_EXFILTRATION
• failed_tcp_ports >= 3 ⇒ PORT_SCAN
• ping_latency > 120 ms ⇒ NET-WORK_CONGESTION
• ping=0 & tcp=0 ⇒ HOST_DOWN
Os tres classificadores sao consumidos apenas por
GET /api/anomalies/classify (fora do loop).